Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	*Biopolymers for long-term use with controlled end-of-life*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (x), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Chemicals		*Energy & Resources*	
	Medium Category	Small Category	Medium Category	Small Category
	Polymeric Materials	*Materials for Energy/Environmental Industries*	*Industrial Biotechnology*	*Bio-plastics*
Concept of Proposed Technology	*Core concept: Design additive systems for biopolymers (PLA/PBS/PBSA and related biopolyesters) that deliver (i) long-term hydrolysis stabilization via synergistic inhibitor + acid-scavenger packages and (ii) on-demand end-of-life (EoL) acceleration under acidic conditions using switchable trigger additives. Integrate bio-based, phosphorus-containing flame-retardant (FR) options for energy/built-environment use.* *Main functions:* • *Service-life extension: minimize chain scission and MVR drift in humid/thermal environments; tune crystallinity to raise service temperature.* • *EoL control: accelerate hydrolysis only under moisture/compost conditions; preserve processing/usage stability in dry/aprotic conditions.* • *Optional FR: bio-based phosphorus cellulose esters or lignin derivatives for reduced*			

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	smoke/toxicity and char promotion. • *Optional UV-stability: introduce UV-stabilizer packages for outdoor long-term stability* *Potential uses:* • *Construction/PV components (e.g., backsheet/encapsulant films and peripheral elements).* • *Construction films, such as vapor barrier film, covering film or diffusion-open film* • *Wall/panel systems and building components requiring predictable service life and controlled EoL.* • *Additional durable biopolymer applications in appliances/interiors where VOC/odor limits apply.*
Rationale for Joint R&D Project	*Industry need: Biopolymers often suffer rapid hydrolytic degradation and limited service temperature. Markets (PV periphery, construction) need durable biobased materials with certifiable reliability and controllable EoL. Integrating stabilization and EoL triggers without processing trade-offs requires mechanism-led design and pilot validation.* *Collaboration rationale:* • *Research institute (Fraunhofer LBF): additive science, synergistic inhibitor + acid scavenger systems; controlled hydrolysis triggers; bio-based FRs; analytics (MVR/DSC/DMTA, migration/VOC/fog), PV/build reliability testing methods, UV-stabilization and testing, LCA/ESG.* • *Korean SMEs: biopolymer compounding scale-up, film/panel processing, testing with local standards, rapid commercialization in domestic supply chains.* • *Joint R&D aligns formulation-to-certification and accelerates market entry in New & Renewable Energy and built-environment segments.*
Key Words	1. *Controlled end-of-life biopolymers* 2. *Hydrolysis stabilization (inhibitor + acid scavenger synergy)* 3. *Bio-based phosphorus flame retardants* 4. *demonstration panels* 5. *Triggered hydrolysis under protic conditions*

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4-5)	End of the Project	Level (6-7)
R&D Objectives	*Stabilization modules:* • *Carbodiimide-based hydrolysis inhibitors coordinated with acid scavengers (e.g., hydroxycarboxylate salts) for synergistic pH regulation and extended durability (per Fraunhofer LBF evidence).* • *Crystallization tuning to raise service temperature (e.g., enabling 100°C service for semi-crystalline PLA).* *EoL trigger modules:* • *Polyol + organic phosphorus compounds (WO2020148089) and hydroxycarboxylate salt + phosphorus systems (WO2021213939/WO2021214207) that remain inert during processing but accelerate hydrolysis under moisture/compost (protic) conditions.* *Bio-based FR options:* • *Phosphorus-containing cellulose esters (WO2021048335) or lignin derivatives to reduce smoke/PHRR and support char formation with low migration.* *UV-stabilization options:* • *Light stabilizers in combination with UV absorbers as additives to improve the long-term stability in outdoor applications.* *Delivery forms and processing:* • *Masterbatch routes, low-VOC designs, migration control; film extrusion and panel fabrication; recyclability-friendly mono-material strategies where possible.*			
R&D Target Specifications	*Hydrolytic stability (water aging):* • *MVR drift ≤10% after 8–12 weeks at 60°C water immersion (vs. rapid drift in non-stabilized PLA within days).* • *Synergy effect: extend stabilization window by ≥5 weeks vs. inhibitor-only reference at equal total additive loading (LBF evidence).* *Thermal aging:* • *At 100°C dry aging, ΔMVR ≤15% after 2000 h for stabilized formulations; non-stabilized PLA shows marked drift from ~600 h (LBF evidence).* *Service temperature and crystallinity:* • *Enable semi-crystalline PLA service temp ~100°C (vs. ~60°C amorphous), while retaining mechanicals (≥90% tensile/impact vs. baseline at target crystallinity).* *EoL trigger performance:* • *Compost/humidity half-life reduction ≥50% vs. non-trigger control; weight loss ≥30% within 6–8 weeks under standardized compost (58°C) or humidity cycling protocols, with no processing degradation in dry/aprotic conditions.* *Emissions/migration:* • *VOC (VDA 278) ≤200 μg/g; Fog (DIN 75201-B) ≤3 mg; extractables ≤0.5% (7 d/40°C) in*			

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R&D Plan

relevant media.

Fire behavior (if applicable):

- *UL 94 VTM-0 for thin films and bulk materials (as applicable); IEC 60695 screening; DIN 4102 B2 (or EN 13501 class target per application).*

Approach

- *Mechanism-led design of stabilization and EoL trigger systems that remain neutral in processing but act under protic service/end-of-life conditions; integrate bio-based flame retardants and/or UV stabilizers for construction materials; validate with reliability and fire standards; assess LCA/ESG.*

Methodologies

- *Analytics: MVR (melt flow), DSC/DMTA, rheology; hydrolysis aging (60–85°C water), thermal aging (100°C), crystallization/optics; VOC/fog/migration.*
- *Fire/reliability: UL 94 VTM/IEC 60695; cone calorimetry (ISO 5660) for FR variants; PV IEC 61215/61730 stress tests; building panel tests (DIN 4102).*
- *Processing & scale-up: twin-screw compounding, film extrusion, lamination; panel fabrication.*
- *LCA/ESG: CO2, resource inputs, end-of-life scenarios.*

Milestones & timeline (3 years)

Phase	**Months**	**Milestone**	**Description**
I	0–2	M1	Kick-off; target components/specs; compliance plan
I	0–6	M2	Stabilization + EoL trigger screening; Top-10 formulations meeting MVR/aging pre-screens
II	7–12	M3	Masterbatch routes; crystallization tuning; initial FR integration; VOC/fog/migration pass
II	13–18	M4	PV/backsheet/encapsulant or various film prototypes; building panels prototypes
II	19–24	M5	Reliability/fire tests passed to target levels; EoL trigger validation (compost/humidity cycles)
II	25–30	M6	Scale-up process windows; LCA/ESG report; cost/ performance model
II	31–36	M7	Certification-ready datasets; tech transfer package; commercialization roadmap

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• *Technical contributions: formulation design (inhibitor + acid scavenger synergy; EoL triggers; bio-based FR); analytics (MVR/DSC/DMTA; migration/VOC/fog); hydrolysis/thermal aging; PV/build reliability and fire pre-screens; UV stabilization and testing; LCA/ESG.* • *Research scope: PLA/PBS/PBSA and related biopolyesters; crystallization tuning; masterbatch and film/panel processing support.* • *Resources: compounding/film labs; analytics/test facilities; project/data management; documentation for Phase II and certification.*
Expected R&R of Korean SME Partner	• *Technical contributions: biopolymer compounding scale-up; film and panel manufacturing; processing window optimization.* • *Development tasks: local PV/build reliability and fire testing coordination; emissions/migration per domestic protocols; cost and supply-chain modeling.* • *Testing/validation: assemble data for Phase I/II reviews; run IEC/DIN/UL tests with Korean test organizations; application-specific validation.* • *Resource commitments: pilot lines, materials, regulatory liaison (K-REACH); participation in tech transfer and commercialization.*

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	• *Technological achievements: durable biopolymer formulations with extended service life (MVR stability in hydrolytic/thermal aging), predictable EoL via protic triggers, and optional bio-based FR performance suited for PV periphery and building panels.* • *Commercialization potential: masterbatch/compound products for PV/built-environment markets; certification-ready datasets; domestic scale-up with Korean SMEs; joint IP and licensing routes.* • *Industrial/economic impact: broadened use of bioplastics in long-life applications; controlled end-of-life to support circularity; reduced smoke/toxic emissions vs. halogen systems; strengthened KR–DE collaboration in New & Renewable Energy and construction sectors.*

6. References

1) Fraunhofer LBF (Kunststoffe 12/2025): “Increasing long-term stability of biopolymers via synergistic additive systems” (MVR stabilization, synergy and crystallization effects).

2) WO2020148089A1: Polyol + phosphorus additive systems for on-demand hydrolysis acceleration.

3) WO2021213939A1; WO2021214207A1: Hydroxycarboxylate salt + phosphorus systems; stabilization and controlled hydrolysis.

4) WO2021048335A1: Phosphorus-containing cellulose esters as bio-based flame retardants.

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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